Mathematical Statistics

Hypothesis Testing

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Hypothesis Testing

- Definition: A hypothesis is a statement about a population paraemter
- Definition: The two complementary hypotheses in a hypothesis testing problem are called the null hypothesis and alternative hypothesis, denoted by H_0 and H_1, respectively.
- Let θ denote a population parameter.
- The general null and alternative hypotheses is $H_0 : \theta \in \Omega_0$ and $H_1 : \theta \in \Omega_0^c$, where Ω_0 is a some subset of the parameter space, and Ω_0^c is its complement.

Examples

- Let θ denote the average change in a patient's blood pressure after taking a drug, and we are interested in testing that the drug has some effect:

$$H_0 : \theta = 0 \text{ versus } H_1 : \theta \neq 0$$

- Let θ denote the proportion of defective items produced by a supplier, and we are interested in whether the supplier pass the quality control:

$$H_0 : \theta \geq \theta_0 \text{ versus } H_1 : \theta < \theta_0,$$

where θ_0 is the maximum acceptable proportion of defective items.

Hypothesis Test

- A hypothesis testing procedure or hypothesis test is a rule that specifies
  - For which sample values the decision is made to accept H_0 as true.
  - For which sample values H_0 is rejected and H_1 is accepted as true.
- The set of the sample space for which H_0 will be rejected is called rejection region or critical region. The complement of the rejection region is called the acceptance region.
- Distinction between "rejecting H_0" and "accepting H_1: the first statement does not imply that what state the experimenter is accepting and only the state defined by H_0 is being rejected.

Type I and II Errors

- In the hypothesis testing, the question is not whether H_0 is true or false.
- The question is whether there is sufficient evidence to reject H_0, much like a court case.
- Our possible actions are: reject H_0 or retain (don?t reject) H_0.

	Decision	
	Retain H_0	Reject H_0
H_0 true	✓	Type I error (false positive)
H_1 true	Type II error (false negative)	✓

Constructing Tests

- Let $X_1, \ldots, X_n \sim f(x \mid \theta)$.
- Hypothesis testing involves the following steps:
  - Choose a test statistic $T_n = T_n(X_1, \ldots, X_n)$.
  - Choose a rejection region $\mathcal{R}$.
  - If $T \in \mathcal{R}$, we reject H_0, otherwise we retain H_0.

Error Rates and Power

- Let $\mathcal{R}$ be the rejection region for H_0. We define the power function by

$$\beta(\theta) = P_\theta(T_n(X_1, \ldots, X_n) \in \mathcal{R}).$$

- We like to construct $\mathcal{R}$ so that $\beta(\theta)$ is small when $\theta \in H_0$ and to be large when $\theta \in H_1$.

Error Rates and Power

The general strategy is:

- Fix α (e.g. $\alpha = 0.05$).
- Try to maximize $\beta(\theta)$ for $\theta \in H_1$ subject to $\beta(\theta) \leq \alpha$ for $\theta \in H_0$.
- A test is size α if

$$sup_{\theta \in \Theta_0} \beta(\theta) = \alpha.$$

- A test is level α if

$$sup_{\theta \in \Theta_0} \beta(\theta) \leq \alpha.$$

- A test with power function $\beta(\theta)$ is unbiased if $\beta(\theta') \geq \beta(\theta'')$ for every $\theta' \in \Omega_1$ and $\theta'' \in \Omega_0$.

Example

- Let $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$ with σ^2 known.
- The following one-sided null and alternative:

$$H_0 : \theta = \theta_0 \text{ versus } H_1 : \theta > \theta_0.$$

- We reject H_0 if $T_n > c$, where $T_n = \frac{\bar{X}_n - \theta_0}{\sigma/\sqrt{n}}$.
- Thus,

$$\beta(\theta) = P_\theta(\frac{\bar{X}_n - \theta_0}{\sigma/\sqrt{n}} > c) = P_\theta(\frac{\bar{X}_n - \theta}{\sigma/\sqrt{n}} > c + \frac{\theta_0 - \theta}{\sigma/\sqrt{n}}) =$$

$$1 - \Phi(c + \frac{\theta_0 - \theta}{\sigma/\sqrt{n}})$$

  where Φ is the cdf of the standard Normal distribution.
- Since $sup_{\theta \in \Theta_0} \beta(\theta) = \beta(\theta_0) = 1 - \Phi(c)$, to get a size α test, we set $1 - \Phi(c) = \alpha$ so that $c = z_{1-\alpha}$, where $Z_\alpha = \Phi^{-1}(\alpha)$.

Methods for Constructing Tests

- Likelihood Ratio Test
- Bayesian Tests
- Union-Intersection and Intersection-Union tests
- The Neyman-Pearson Test
- The Wald test and Score test
- The permutation test.

Likelihood Ratio Tests

- Let $X_1, \ldots, X_n$ be a random sample from a population with pdf or pmf $f(x \mid \theta)$, where θ is a k-dimensional vector.
- The likelihood function:

$$L(\theta \mid x_1, \ldots, x_n) = L(\theta \mid \mathbf{x}) = f(\mathbf{x} \mid \theta) = \prod_{i=1}^{n} f(x_i \mid \theta).$$

- Let Θ denote the entire parameter space.
- Likelihood ratio test statistic for testing $H_0 : \theta \in \Theta_0$ versus $H_1; \theta \in \Theta_1$ is

$$\lambda(\mathbf{x}) = \frac{sup_{\theta \in \Theta_0} L(\theta \mid \mathbf{x})}{sup_{\theta \in \Theta} L(\theta \mid \mathbf{x})}.$$

- This test rejects H_0 if $\lambda(\mathbf{x}) \leq c$.

Interpretation

The rational behind LTRs may best be understood in the situation in which $f(x \mid \theta)$ is the pmf of a discrete random variable.

- The numerator of $\lambda(\boldsymbol{x})$ is the maximum probability of the observed sample, where the maximum is computed over the parameters in the null hypothesis.
- The denominator of $\lambda(\boldsymbol{x})$ is the maximum probability of the observed sample, where the maximum is computed over all possible parameters.
- The ratio of these two maxima is small if there are parameters points in the alternative hypothesis for which the observed sample is much more likely than for any parameter points in the null hypothesis.

Normal Example

- $X_1, \ldots, X_n \sim N(\mu, 1)$
- $$H_0 : \theta = \theta_0 \text{ versus } H_1 : \theta \neq \theta_0.$$
- After some algebra, we can show that
$$\lambda(X) = exp(-\frac{n}{2}(\bar{X} - \theta_0)^2),$$
by noting
$$\sum_{i=1}^{n}(X_i - \theta_0)^2 = \sum_{i=1}^{n}(X_i - \bar{X})^2 + \sum_{i=1}^{n}(\bar{X} - \theta_0)^2.$$
- Thus,
$$R = \{x : \lambda(x) \leq c\} = \{x :\mid \bar{x} - \theta_0 \mid \geq c'\},$$
where $c' = \sqrt{-2logc/n}$.
- Choosing c' to makes the test level α: reject H_0 if $\mid T_n \mid > z_{1-\alpha/2}$, where $T_n = \sqrt{n}(\bar{X} - \theta_0)$.

Example 8.2.3

- Let $X_1, \ldots, X_n$ be a random sample from an exponential population with pdf,

$$f(x \mid \theta) = \begin{cases} e^{-(x-\theta)} & x \geq \theta \\ 0 & x < \theta \end{cases}$$

  where $\infty < \theta < \infty$

- The likelihood function is

$$L(\theta \mid x) = \begin{cases} e^{-\sum_i x_i + n\theta} & \theta \leq x_{(1)} \\ 0 & \theta > x_{(1)} \end{cases}$$

- Consider testing $H_0 : \theta \leq \theta_0$ versus $H_1 : \theta > \theta_0$.
- $L(\theta \mid x)$ is an increasing function of θ on $-\infty < \theta \leq x_{(1)}$. Thus, the. denominator of $\lambda(x)$, the unrestricted maximum of $L(\theta \mid x)$, is $L(x_{(1)} \mid x) = e^{-\sum_i x_i + n x_{(1)}}$.

Example 8.2.3

- If $x_{(1)} \leq \theta_0$, the numerator of $\lambda(x)$ is also $L(x_{(1)} \mid x)$.
- If $x_{(1)} > \theta_0$, since we are maximizing $L(\theta \mid x)$ over $\theta \leq \theta_0$, the numerator of $\lambda(x)$ is $L(\theta_0 \mid x)$.
- Therefore, the likelihood ratio test statistic is

$$\lambda(x) = \begin{cases} e^{-n(x_{(1)}-\theta_0)} & x_{(1)} > \theta_0 \\ 1 & x_{(1)} \leq \theta_0 \end{cases}$$

- An LRT, a test statistic that rejects H_0 if $\lambda(x) \leq c$, is a test with rejection region $\{x : x_{(1)} \geq \theta_0 - \frac{\log c}{n}\}$

LRTs and Sufficiency

- If $T(X)$ is a sufficient statistic for θ with pdf or pmf $g(t \mid \theta)$, we may consider construction of an LRT, based on T and its likelihood function $L^*(\theta \mid t) = g(t \mid \theta)$, rather than on the sample X and its likelihood function $L(\theta \mid x)$.
- Theorem: If $T(X)$ is a sufficient statistic for θ and $\lambda^*(t)$ and $\lambda(x)$ are the LRT statistics based on T and X, respectively, then $\lambda^*(T(x)) = \lambda(x)$ for every x in the sample space.

Proof

- Proof: from the Factorization Theorem, we have that the pdf or pmf of X can be written as $f(x \mid \theta) = g(T(x) \mid \theta)h(x)$ where $g(t \mid \theta)$ is the pdf or pmf of T. Thus,

$$\lambda(x) = \frac{\sup_{\Omega_0} L(\theta \mid x)}{\sup_{\Omega} L(\theta \mid x)} = \frac{\sup_{\Omega_0} f(\theta \mid x)}{\sup_{\Omega} f(\theta \mid x)} = \frac{\sup_{\Omega_0} g(T(x) \mid \theta)h(x)}{\sup_{\Omega} g(T(x) \mid \theta)h(x)}$$

$$= \frac{\sup_{\Omega_0} g(T(x) \mid \theta)}{\sup_{\Omega} g(T(x) \mid \theta)} = \frac{\sup_{\Omega_0} L^*(\theta \mid T(x)}{\sup_{\Omega} L^*(\theta \mid T(x)} = \lambda^*(T(x))$$

Example 8.2.5

- Let $X_1, \ldots, X_n$ be a random sample from an exponential population with pdf,

$$f(x \mid \theta) = \begin{cases} e^{-(x-\theta)} & x \geq \theta \\ 0 & x < \theta \end{cases}$$

  where $\infty < \theta < \infty$
- The likelihood function is

$$L(\theta \mid x) = \begin{cases} e^{-\sum_i x_i + n\theta} & \theta \leq x_{(1)} \\ 0 & \theta > x_{(1)} \end{cases}$$

- . $L(\theta \mid x)$ is an increasing function of θ on $-\infty < \theta \leq x_{(1)}$

Example 8.2.5

- Hence $X_{(1)}$ is the sufficient statistics, and the likelihood function for $X_{(1)}$ is:

$$L^*(\theta \mid x_{(1)}) = \begin{cases} ne^{-n(x_{(1)}-\theta)} & \theta \leq x_{(1)} \\ 0 & \theta > x_{(1)} \end{cases}$$

- Hence, this likelihood could be used to derive the likelihood ratio test of $H_0 : \theta \leq \theta_0$ versus $H_1; \theta > \theta_0$.

LRTs

- Likelihood ratio tests are also useful in situations where there are nuisance parameters, which are parameters that are present in a model but are not of direct inferential interest.
- The presence of such nuisance parameters does not affect the LRT construction but might lead to a different test.

Example 8.2.6

- Let $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$.
- $$H_0 : \mu \leq \mu_0 \text{ versus } H_1 : \mu > \mu_0$$
- Let $\widehat{\mu}$ and $\widehat{\sigma}^2$ be the MLEs of μ and σ^2 without any restriction and $\widehat{\mu}_0$ and $\widehat{\sigma}_0^2$ be the MLE of μ and σ^2 under H_0.
- If $\widehat{\mu} \leq \mu_0$, $\widehat{\mu} = \widehat{\mu}_0$ and $\widehat{\sigma}^2 = \widehat{\sigma}_0^2$, and hence $\lambda(x) = 1$.
- If $\widehat{\mu} > \mu_0$, the likelihood ratio test statistic is:

$$\lambda(x_1, \ldots, x_n) = \frac{L(\mu_0, \widehat{\sigma}_0)}{L(\widehat{\mu}, \widehat{\sigma})}.$$

Example 8.2.6

- We can show that $\lambda(x_1, \ldots, x_n) < c$ corresponds to rejecting when $T_n > k$ for some constant k, where

$$T_n = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}.$$

  where $S = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X})^2$.
- Under H_0, T_n has a t-distribution with $n-1$ degrees of freedom. Hence, the final test is: reject H_0 if $T_n > t_{n-1,\alpha/2}$.
- This is called Student's t-test and was invented by William Gosset working at Guiness Breweries and writing under the pseudonym Student

Definition of Uniformly Most Powerful (UMP) Test

Let $\mathcal{C}_\alpha$ denote all levels α tests. A test in $\mathcal{C}_\alpha$ with the power function $\beta(.)$ is called uniformly most powerful (UMP) test if the following holds: if $\beta'(.)$ is a power function of any other test in $\mathcal{C}_\alpha$ then $\beta'(\theta) \leq \beta(\theta)$ for all $\theta \in \Theta_1$.

Neyman-Pearson Lemma

- Let $f(. \mid \theta)$ be the pdf or pmf of the data $X = (X_1, \ldots, X_n)$
- Consider a simple null and alternative hypotheses $H_0 : \theta = \theta_0$ versus $H_1 : \theta = \theta_1$.
- Using a test with rejection region R that satisfies (8.3.1)$x \in R$ if $f(x \mid \theta_1) > kf(x \mid \theta_0)$; $x \in R^c$ if $f(x \mid \theta_1) < kf(x \mid \theta_0)$ for some $k \leq 0$ and (8.3.2)$\alpha = P_{\theta_0}(X \in R)$ Then,
  - a. (Sufficiency) Any test that satisfies (8.3.1) and (8.3.2) is a UMP level α test.
  - b. (Necessity) If there exists a test satisfying (8.3.1) and (8.3.2) with $k > 0$, then every UMP level α test is a size α test (satisfies (8.3.2)) and everyone UMP level α test satisfies (8.3.1) except perhaps on a set A satisfying $P_{\theta_0}(X \in A) - P_{\theta_1}(X \in A) = 0$

Proof

We will prove the theorem for the case that $f(x\,|_0)$ and $f(x \mid \theta_1)$ are continuous random variables.

- First note that the above defined test is a size α test. To ease notation, we define a test function on the sample space, which is 1 if $x \in R$ and 0 if $x \in R^c$.
- Let $\phi(x)$ be the test function of a test satisfying the above conditions. That is, $\phi(x) = 1$ if $f(x \mid \theta_1) > kf(x \mid \theta_0)$, and $\phi(x) = 0$ if $f(x \mid \theta_1) < kf(x \mid \theta_0)$.
- Let $\phi'(x)$ be the test function of any other level α test, and let $\beta(\theta)$ and $\beta'(\theta)$ be the power functions corresponding to the tests ϕ and ϕ', respectively.
- Since $0 \leq \phi'(x) \leq 1$, by the definition of ϕ, we obtain that

$$(\phi(x) - \phi'(x))(f(x \mid \theta_1) - kf(x \mid \theta_0)) \geq 0 \text{ for every } x$$

Proof

- Thus,

$$(8.3.3)\ 0 \leq \int [\phi(x) - \phi'(x)][f(x \mid \theta_1) - kf(x \mid \theta_0)]dx =$$

$$\beta(\theta_1) - \beta'(\theta_1) - k(\beta(\theta_0) - \beta'(\theta_0))$$

- Statement (a) is shown by noting that ϕ is a size α test and ϕ' is a level α teat, which implies that $\beta(\theta_0) - \beta'(\theta_0) = \alpha - \beta'(\theta_0) \geq 0$. Using (8.3.3) and the fact $k \geq 0$, we obtain that $0 \leq \beta(\theta_1) - \beta'(\theta_1)$, which implies that $\beta(\theta_1) \geq \beta'(\theta_1)$. Thus, ϕ is a UMP size α test.

Proof

- To show statement (b), let ϕ' is a test function for any UMP level α test. By part (a), ϕ, satisfying (8.3.1) and (8.3.2) is also a UMP level α test, thus $\beta(\theta_1) = \beta'(\theta_1)$. This fact, (8.3.3), and $k \geq 0$ implies that

$$\alpha - \beta'(\theta_0) = \beta(\theta_0) - \beta'(\theta_0) \leq 0.$$

  Since ϕ' is a level α test, $\beta'(\theta_0) \leq \alpha$. Thus, $\beta'(\theta_0) = \alpha$, which implies that (8.3.3) is an equality in this case. But, the nonnegative integrand $[\phi(x) - \phi'(x)][f(x \mid \theta_1) - kf(x \mid \theta_0)]$ will have a zero integral only if ϕ' satisfies (8.3.1), except perhaps on a set A with $\int_A f(x \mid \theta_i)dx = 0, i = 0, 1$. This implies that the last result in statement (b) is true.

If $f(x \mid \theta_0)$ is not zero, it may be easier to use $f(x \mid \theta_1)/f(x \mid \theta_0) > k$.

Neyman-Pearson and Sufficiency

Corollary 8.3.13:

- Suppose that $T(X)$ is a sufficient statistic for θ and $g(t \mid \theta_i)$ is the pdf or pmf of T corresponding to θ_i, $i = 0, 1$.
- Then, any test based on T with rejection region S is a UMP level α test if and only if it satisfies that

. (8.3.4) $t \in S$ if $g(t \mid \theta_1) > kg(t \mid \theta_0)$ and

$t \in S^c$ if $g(t \mid \theta_1) < kg(t \mid \theta_0)$,

for some $k \geq 0$, where

(8.3.5) $\alpha = P_{\theta_0}(T \in S)$.

Proof

- For the original sample X, the test based on T has the rejection region $R = \{x : T(x) \in S\}$.
- By the Factorization Theorem, the pdf or pmf of X can be written as $f(x \mid \theta_i) = g(T(x) \mid \theta_i)h(x), i = 0, 1$, for some nonnegative function $h(x)$.
- Multiplying the inequalities in (8.3.4) by this nonnegative function, we see that R satisfies

$$x \in R \text{ if } f(x \mid \theta_1) = g(T(x) \mid \theta_1)h(x) >$$

$$kg(T(x) \mid \theta_0)h(x) = kf(x \mid \theta_0)$$

and

$$x \in R^c \text{ if } f(x \mid \theta_1) = g(T(x) \mid \theta_1)h(x) <$$

$$kg(T(x) \mid \theta_0)h(x) = kf(x \mid \theta_0)$$

Proof

- Also, by (8.3.5), we obtain that

$$P_{\theta_0}(X \in R) = P_{\theta_0}(T(X) \in S) = \alpha.$$

- So, by the sufficiency part of the Neyman-Pearson Lemma, the test based on $T$is a UMP level α test.
- Also, by (8.3.5), we obtain that

$$P_{\theta_0}(X \in R) = P_{\theta_0}(T(X) \in S) = \alpha.$$

- So, by the sufficiency part of the Neyman-Pearson Lemma, the test based on $T$is a UMP level α test.

UMP binomial test

- Let $X \sim \textit{binomial}(2, \theta)$. Want to test $H_0 : \theta = 1/2$ versus $H_1 : \theta = 3/4$.
- The ratios of the pmfs gives us

$$\frac{f(0 \mid \theta = 3/4)}{f(0 \mid \theta = 1/2)} = 1/4, \frac{f(1 \mid \theta = 3/4)}{f(0 \mid \theta = 1/2)} = 3/4,$$

$$\frac{f(2 \mid \theta = 3/4)}{f(0 \mid \theta = 1/2)} = 9/4;$$

- If we choose $3/4 < k < 9/4$, the Neyman-Pearson Lemma says that rejects H_0 if $X = 2$ $(\frac{f(x|\theta=3/4)}{f(x|\theta=1/2)} > k)$ is the UMP level $\alpha = P(X = 2 \mid \theta = 1/2) = 1/4$ test.
- If we choose $1/4 < k < 3/4$, the Neyman-Pearson Lemma says that rejects H_0 if $X = 1$, or 2 $(\frac{f(x|\theta=3/4)}{f(x|\theta=1/2)} > k)$ is the UMP level $\alpha = P(X = 1, \text{ or } 2 \mid \theta = 1/2) = 3/4$ test.

UMP binomial test

- If we choose $k < 1/4$ or $k > 9/4$, We obtain the UMP level $\alpha = 1$ or $\alpha = 0$ test.
- If $k = 3/4$, the UMP test (8.3.1) says that we reject H_0 if $x = 2$ and accept H_0 for $x = 0$ but leaves our decision for $x = 1$ undetermined.
- But, we accept H_0 for $x = 1$, we get the UMP level $\alpha = 1/4$. If we reject H_0 for $x = 1$,we get the UMP test $\alpha = 3/4$.
- This example shows that for a discrete distribution, the α level at which a test can be done is a function of the particular pmf.

Extension of Neyman-Pearson Test

- We now extend Neyman-Pearson result to result to a class of parametric problems in which the likelihood functions have a special property.
- Monotone likelihood ratio (definition): For the distribution of X is in $\mathcal{P} = \{P_\theta, \theta \in \Omega\}$, a parametric family indexed by a real-valued θ, and and that $\mathcal{P}$ is dominated by a σ-finite measure v. Let $f_\theta = dP_\theta/dv$.
- The family $\mathcal{P}$ is said to have monotone likelihood ratio in $T(X)$ (a real-valued statistic) if and only if, for any $\theta_1 < \theta_2$, $f_{\theta_2}(x)/f_{\theta_1}(x)$ is a nondecreasing function of $T(x)$ for values x at which at least one of $f_{\theta_1}(x)$ and $f_{\theta_2}(x)$ is positive. Note that $c/0$ is defined as ∞ if $c > 0$.

Extension of Neyman-Pearson Test

Theorem (Karlin-Rubin):

- Suppose that X has a distribution in in $\mathcal{P} = \{P_\theta, \theta \in \Omega\}$ that has monotone likelihood ratio in a sufficient statistic $T(X)$ for θ.
- The problem of testing $H_0 : \theta \leq \theta_0$ versus $H_1 : \theta > \theta_0$.
- Then, for any t_0, the test that reject H_0 if and only if $T > t_0$ is a UMP level α test, where $\alpha = P_{\theta_0}(T > t_0)$.

Proof

- Let $\beta(\theta) = P_\theta(T > t_0)$ be the power function of the test. For $\theta' > \theta_0$, consider testing $H_0' : \theta = \theta_0$ versus $H_1' : \theta = \theta'$.
- Since the family of pdfs or pmfs of T has an MLR, $\beta(\theta)$ is nondecreasing (Homework- Exercise 8.34). So
- (i) $sup_{\theta \leq \theta_0} \beta(\theta) = \beta(\theta_0) = \alpha$,and this is a size α test
- (ii) If we define

$$k' = inf_{t \in \mathcal{T}} \frac{g(t \mid \theta')}{g(t \mid \theta_0)},$$

where $\mathcal{T} = \{t : t > t_0, g(t \mid \theta') > 0 \text{ or } g(t \mid \theta_0) > 0\}$. Then, we obtain that

$$T > t_0 \iff \frac{g(t \mid \theta')}{g(t \mid \theta_0)} > k'.$$

Proof, continued

- Together with Corollary 8.3.13, (i) and (ii) imply that $\beta(\theta') \geq \beta^*(\theta')$, where $\beta^*(\theta)$ is the power function for any other level α test of H_0'; that is, any test. satisfying $\beta(\theta_0)^* \leq \alpha$.
- However, any level α test of H_0 satisfies $\beta^*(\theta_0) \leq sup_{\theta \in \Omega_0} \beta^*(\theta) \leq \alpha$. Thus, by Neyman-Pearson Lemma, we have that $\beta(\theta') \geq \beta(\theta')^*$. Since θ' was arbitrary, the test is a UMP level α test.

Similar Result

- By analogous argument, it can be shown that under the conditions of Theorem 8.3.17, the test that reject $H_0 : \theta \geq \theta_0$ in favor of $H_1 : \theta < \theta_0$ if and only if $T < t_0$ is a UMP level $\alpha = P_{\theta_0}(T < t_0)$.
- Although we prefer to use a UMP level α test, unfortunately for most problems, there is no UMP level α test.

Example 8.3.18

- $X_1, \ldots, X_n \sim N(\theta, \sigma^2)$, where σ is known.
- Consider $H_0 : \theta \geq \theta_0$ versus $H_1 : \theta < \theta_0$ using the test that reject H_0 if

$$\bar{X} < -\frac{\sigma z_\alpha}{\sqrt{n}} + \theta_0$$

- As $\bar{X}$ is a sufficient statistics and its distribution has an MLR, it follows from Theorem 8.3.17 that the test is a UMP level α test.
- The power function,

$$\beta(\theta) = P_\theta(\bar{X} < -\frac{\sigma z_\alpha}{\sqrt{n}} + \theta_0),$$

is a decreasing function of θ, the value of α is given by $sup_{\theta \geq \theta_0} \beta(\theta) = \beta(\theta_0)$.

Nonexistence of UMP test

- Although we would prefer to use a UMP level α test, unfortunately, for many problems there is no UMP level α test because the class of level α test is so large that no one test dominates all the others in terms of power.
- One solution is to restrict the UMP to the class of unbiased estimators.

Nonexistence of UMP test

Example 8.3.19:

- $X_1, \ldots, X_n \sim N(\theta, \sigma^2)$, where σ is known.
- Consider $H_0 : \theta = \theta_0$ versus $H_0 : \theta \neq \theta_0$.
- A level α testis any test that satisfies

$$(8.3.6)\ P_{\theta_0}(\text{reject } H_0) \leq \alpha$$

- Consider an alternative parameter point $\theta_1 < \theta_0$. From the result in Example 8.3.18, it follows that among all tests that satisfy (8.3.6), the test that rejects H_0 if $\bar{X} < -\sigma z_\alpha/\sqrt{n} + \theta_0$ has the highest possible power at θ_1. Call this Test 1.

Nonexistence of UMP test

- By part (b) (necessity) of the Newman-Pearson Lemma, any other level α test that has as high a power as Test 1 at θ_1 must have the same rejection region as Test 1 except possibly for a set A satisfying $\int_A f(x \mid \theta_i)dx = 0$, $i = 0, 1$.
- Thus, if a UMP level α test exists for this problem, it must be Test 1 because no other test has as high a power as Test 1 at θ_1.

Nonexistence of UMP test

- Now consider Test 2, which rejects H_0 if $\bar{X} > \sigma z_\alpha/\sqrt{n} + \theta_0$, which is a level α test.
- Let $\beta_i(\theta)$ be the power function of Test i.
- For any $\theta_2 > \theta_0$,

$$\beta_2(\theta_2) = P_{\theta_2}(\bar{X} > \frac{\sigma z_\alpha}{\sqrt{n}} + \theta_0)$$

$$= P_{\theta_2}(\frac{\bar{X} - \theta_2}{\sigma\sqrt{n}} > z_\alpha + \frac{\theta_0 - \theta_2}{\sigma\sqrt{n}})$$

$$> P(Z > z_\alpha) = P(Z < -z_\alpha),$$

since $\theta_0 - \theta_2 < 0$, where $Z \sim N(0, 1)$,

$$> P_{\theta_2}(\frac{\bar{X} - \theta_2}{\sigma\sqrt{n}} < -z_\alpha + \frac{\theta_0 - \theta_2}{\sigma\sqrt{n}})$$

$$= P_{\theta_2}(\bar{X} < -\frac{\sigma z_\alpha}{\sqrt{n}} + \theta_0) = \beta_1(\theta_2),$$

Nonexistence of UMP test

- Thus Test 1 is not a UMP level α test because test 2 has a higher power than Test 1 at θ_2
- We showed before that if there were a UMP level α test, it would have to be Test 1.
- Contradiction.

Asymptotic Distribution of Likelihood Ratio Statistic

- Notation: let χ^2_p be a Chi-squared distribution with p degrees of freedom, and let $\chi^2_{p,\alpha}$ be the $\alpha 100\%$ percentile, $P(W < \chi^2_{p,\alpha}) = \alpha$, if $W \sim \chi^2_p$.
- Here $p = dim(\Omega) - dim(\Omega_0)$= the difference in the number of parameters between the two models under Ω and H_0 . The number of free parameters is the dimensionality of the space that the parameter vector can move in.

Asymptotic Distribution of Likelihood Ratio Statistic

- Theorem: Assume the same assumptions as the ones in the theorem for asymptotic normality of the MLE. Under H_0

$$T_n = -2log\lambda(x_1, \ldots, x_n) \xrightarrow{d} \chi^2_r.$$

- Hence, an asymptotically level α test is: reject H_0 when $T_n > \chi^2_{r,1-\alpha}$ where r is the difference between the number of free parameters specified by $\theta \in \Omega_0$ and the number of free parameters specified by $\theta \in \Omega$.

Proof of Asymptotic Result for LRT in a Simple Case

- Consider $H_0 : \theta = \theta_0$. The likleihood ratio test statistics: $W(y) = -2log\lambda(y) = 2(l(\widehat{\theta}) - l(\theta_0))$, where $\lambda(y)$ is the likelihood ratio.
- Using Taylor expansion for $l'(\widehat{\theta}) = 0$ around θ_0, we obtain

$$l'(\widehat{\theta}) - l'(\theta_0) = -l'(\theta_0) = (\widehat{\theta} - \theta_0)\{l''(\theta_0) + o_P(1)\}$$

- Using Taylor expansion for $l(\widehat{\theta})$ around θ_0:

$$l(\widehat{\theta}) - l(\theta_0) = (\widehat{\theta} - \theta_0)l'(\theta_0) + \frac{1}{2}(\widehat{\theta} - \theta_0)^2 l''(\theta_0) + o_P((\widehat{\theta} - \theta_0)^2)$$

Proof of Asymptotic Result for LRT

- Combining those two results, we obtain

$$l(\widehat{\theta}) - l(\theta_0) = -(\widehat{\theta} - \theta_0)^2 l''(\theta_0) + \frac{1}{2}(\widehat{\theta} - \theta_0)^2 l''(\theta_0) + o_P((\widehat{\theta} - \theta_0)^2)$$

- Furthermore,
$l(\widehat{\theta}) - l(\theta_0) = n(\widehat{\theta} - \theta_0)^2 \{-\frac{1}{2n} l''(\theta_0) + o_P(\frac{1}{n})\}$

Proof of Asymptotic Result for LRT

- Some facts: based on the central limit theorem and the law of large numbers, we obtain the following results:

$$\sqrt{n}(\widehat{\theta} - \theta_0) \to N(0, \frac{1}{I(\theta_0)}) \text{ in distribution}$$

$$\frac{1}{n} l''(\theta_0) \to I(\theta_0) \text{ in probability}$$

- Using Slutsky's theorem, we obtain that $W(X) \to \chi_1^2$ in distribution under H_0.

Example 10.3.4 (Multinomial)

- $X_1, \ldots, X_n$ are i.i.d. $P_\theta(X_i = j) = p_j, j = 1, \ldots, 5$. Here $\theta = (p_1, \ldots, p_5)$ $\theta = (p_1, \ldots, p_5)$ y_j is the number of $x_1, \ldots, x_n$ equal to j.
- $H_0 : p_1 = p_2 = p_3, p_4 = p_5$ versus $H_1 : H_0$ is not true
- Since $p_5 = 1 - p_1 - \ldots - p_4$, there are 4 free parameters, $p_1, \ldots, p_4$.
- Note that there is only one free parameter in the set specified by H_0. Thus, $r = 4 - 1 = 3$.
- To compute $\lambda(x)$, the MLE for θ with and without H_0.
- Without H_0, the MLE for θ is: $\widehat{p}_j = y_j/n$, with $n = \sum_{j=1}^{5} y_j$.

Example 10.3.4 (Multinomial), continued

- Under H_0, the likelihood is:

$$L(\theta \mid x) = p_1^{y_1+y_2+y_3}\left(\frac{1-3p_1}{2}\right)^{y_4+y_5}.$$

- Under H_0, setting the derivative of log-likelihood equal to 0 shows that the MLE of p_1 is $\widehat{p}_{10} = (y_1 + y_2 + y_3)/(3n)$ and $\widehat{p}_{20} = \widehat{p}_{30} = \widehat{p}_{10}$.
- Also, $\widehat{p}_{40} = \widehat{p}_{50} = (1 - 3\widehat{p}_{10})/2$
- Thus, the log-likelihood ratio test statistic:

$$\lambda(x) = \left(\frac{y_1 + y_2 + y_3}{3y_1}\right)^{y_1}\left(\frac{y_1 + y_2 + y_3}{3y_2}\right)^{y_2}$$

$$\left(\frac{y_1 + y_2 + y_3}{3y_3}\right)^{y_3}\left(\frac{y_4 + y_5}{2y_4}\right)^{y_4}\left(\frac{y_4 + y_5}{2y_5}\right)^{y_5}$$

Example 10.3.4 (Multinomial), continued

- Thus the test statistic is:

$$W(x) = -2log\lambda(x) = 2\sum_{i=1}^{5} y_i log(\frac{y_i}{m_i}),$$

  where $m_1 = m_2 = m_3 = (y_1 + y_2 + y_3)/3$ and $m_4 = m_5 = (y_4 + y_5)/2$.
- The asymptotic size α test reject H_0 if $-2log\lambda(x) \geq \chi^2_{3,\alpha}$.

Wald Test

- $X_1, \ldots, X_n \sim f(. \mid \theta)$
- Let $\widehat{\theta}$ be the MLE of θ.
- $H_0 : \theta = \theta_0$ versus $H_1 : \theta \neq \theta_0$.
- By the asymptotic theorem for the MLE, we have that the asymptotic variance of $\widehat{\theta}$ is: $1/I(\theta)$, where $I(\theta)$ is the Fisher information matrix for $X = (X_1, \ldots, X_n)$.
- Wald Test Statistic:

$$W_n = \sqrt{I(\widehat{\theta})}(\widehat{\theta} - \theta_0),$$

  which has an asymptotically standard normal distribution under H_0.
- The rejection region: $\mid W_n \mid \geq z_{\alpha/2}$.

Example 10.3.5- Wald test

- $X_1, \ldots, X_n \sim Bernoulli(p)$.
- $H_0 : p = p_0$ versus $H_1 : p \neq p_0$.
- The MLE for p: $\widehat{p}_n = \sum_{i=1}^{n} X_i/n$.
- Wald Test Statistic:

$$W_n = \frac{\widehat{p}_n - p_0}{\sqrt{\widehat{p}_n(1-\widehat{p}_n)/n}} \xrightarrow{d} N(0,1)$$

  under H_0.
- Thus, the large-sample Wald test rejects H_0 if $\mid W_n \mid > z_{\alpha/2}$

Score Test

- $X_1, \ldots, X_n \sim f(. \mid \theta)$
- Let $\widehat{\theta}$ be the MLE of θ.
- $H_0 : \theta = \theta_0$ versus $H_1 : \theta \neq \theta_0$.
- By the asymptotic theorem for the MLE, we have that the asymptotic variance of $\widehat{\theta}$ is: $1/I(\theta)$, where $I(\theta)$ is the Fisher information matrix for $X = (X_1, \ldots, X_n)$.
- Score Test Statistic:

$$R_n = \sqrt{I(\theta_0)}(\widehat{\theta} - \theta_0),$$

  which has an asymptotically standard normal distribution under H_0.
- The rejection region: $\mid R_n \mid \geq z_{\alpha/2}$.

Example 10.3.5, Score Test

- $X_1, \ldots, X_n \sim Bernoulli(p)$.
- $H_0 : p = p_0$ versus $H_1 : p \neq p_0$.
- The MLE for p: $\widehat{p}_n = \sum_{i=1}^{n} X_i/n$, and the variance for $\widehat{p}_n$ is: $p(1-p)/n$.
- $$\frac{\widehat{p}_n - p}{\sqrt{p(1-p)/n}} \xrightarrow{d} N(0,1)$$
- Under H_0;

$$R_n = \frac{\widehat{p}_n - p_0}{\sqrt{p_0(1-p_0)/n}} \xrightarrow{d} N(0,1)$$

- Thus, the large-sample Score test rejects H_0 if $\mid R_n \mid > z_{\alpha/2}$

Comparison of LRT, Wald, and Score Tests

- Score Tests: Fit just the restricted model
- Wald Tests: Fit just the unrestricted model
- Likelihood Ratio Tests: Fit Both

Union-Intersection Tests

- We can express the null hypothesis as follows:

$$H_0 : \theta \in \cap_{\gamma \in \Gamma} \Theta_\gamma$$

  where Γ is an arbitrary index set.
- Suppose that tests are available for each the problems of testing $H_{0\gamma} : \theta \in \Theta_\gamma$ versus $H_{1\gamma} : \theta \in \Theta_\gamma^c$, saying the corresponding rejection region being $\{x : T_\gamma(x) \in R_\gamma\}$
- The rejection region for H_0 is

$$\bigcup \{x : T_\gamma(x) \in R_\gamma\}$$

- The rationale is simple. If any one of the hypotheses $H_{0\gamma}$ is rejected, then H_0 is rejected. Only if each of the hypotheses $H_{0\gamma}$ is accepted as true will the intersection H_0 will be accepted as true.

Intersection-Union Tests

- In some situations, a simple expression for rejection region can be derived.
-
- If the rejection regions of the individual hypotheses are all of the form $\{x : T_\gamma(x) \geq c\}$ (c independent of γ). In such cases, the rejection region ofr H_0 is:

$$\bigcup_{\gamma \in \Gamma} \{x : T_\gamma(x) \geq c\} = \{x : sup_{\gamma \in \Gamma} T_\gamma(x) \geq c\}$$

- Thus, the test statistic for testing H_0 is $T(x) = sup_{\gamma \in \Gamma} T_\gamma(x)$.

Example. 8.2.8

- Let $X_1, \ldots, X_n$ be i.i.d $\sim N(\mu, \sigma^2)$.
- $H_0 : \mu = \mu_0$ versus $H_1 : \mu \neq \mu_0$
- We can write H_0 as the intersection of two sets,

$$H_0 : \{\mu \leq \mu_0\} \cap \{\mu : \mu \geq \mu_0\}.$$

- The LRT of $H_{0L} : \mu \leq \mu_0$ versus $H_{1L} : \mu > \mu_0$ is reject $H_{0L} : \mu \leq \mu_0$ in favor of $H_{1L} : \mu > \mu_0$ if $\frac{\bar{X}-\mu_0}{S/\sqrt{n}} \geq t_L$. Similarly, the LRT of $H_{0U} : \mu \geq \mu_0$ versus $H_{1U} : \mu < \mu_0$ is reject H_{0U} if $\frac{\bar{X}-\mu_0}{S/\sqrt{n}} \leq t_U$.
- Thus the union-intersection test of $H_0 : \mu = \mu_0$ versus $H_1 : \mu \neq \mu_0$ can be formed from these two LRTs: reject H_0 if $\frac{\bar{X}-\mu_0}{S/\sqrt{n}} \geq t_L$ or $\frac{\bar{X}-\mu_0}{S/\sqrt{n}} \leq t_U$.

Intersection-Union Tests

- Th intersection-union test can be useful if we can express the null hypothesis as follows:

$$H_0 : \theta \in \cup_{\gamma \in \Gamma} \Theta_\gamma$$

  where Γ is an arbitrary index set.
- Suppose that tests are available for each the problems of testing $H_{0\gamma} : \theta \in \Theta_\gamma$ versus $H_{1\gamma} : \theta \in \Theta_\gamma^c$, saying the corresponding rejection region being $\{x : T_\gamma(x) \in R_\gamma\}$
- The rejection region for H_0 is

$$\bigcap \{x : T_\gamma(x) \in R_\gamma\}$$

- From H_0, we can see that H_0 is false if and only if all of $H_{0\gamma}$ are false. Hence, we reject H_0 if and only if each of the individual hypotheses $H_{0\gamma}$ can be rejected.

Union-Intersection Tests

- If the rejection regions of the individual hypotheses are all of the form $\{x : T_\gamma(x) \geq c\}$ (c independent of γ). In such cases, the rejection region of H_0 is:

$$\bigcap \{x : T_\gamma(x) \geq c\} = \{x : inf_{\gamma \in \Gamma} T_\gamma(x) \geq c\}$$

Example 8.2.9

- Two important parameters for assessing the quality of upholstery fabric: θ_1, the mean breaking strength; θ_2, the probability of passing a flammability test.
- Standards say that if θ_1 should be over 50 pounds and θ_2 should be over 0.95, the fabric is accepted.
- Our null and alternative hypotheses:

$$H_0 : \{\theta_1 \leq 50 \text{ or } \theta_2 \leq 0.95\}$$

versus

$$H_1 : \{\theta_1 > 50 \text{ and } \theta_2 > 0.95\}$$

- $X_1, \ldots, X_n \sim N(\theta_1, \sigma^2)$: measurements of breaking strengths. $Y_1, \ldots, Y_m \sim Bernoulli(\theta_2)$: results of m flammability tests.

Example 8.2.9

- The LRT of $H_{01} : \theta_1 \leq 50$ will reject H_{01} if $(\bar{X} - 50)/S/\sqrt{n} > t$.
- The LRT will reject $H_{02} : \theta_2 \leq 0.95$ if $\sum_{i=1}^{m} Y_i > b$.
- Put together , the rejection region for the intersection-union test is given by

$$\{(x, y) : \frac{\bar{x} - 50}{s/\sqrt{n}} > t \text{ and } \sum_{i=1}^{m} y_i > b\}$$

P-value

- After a hypothesis test is done, the conclusion must be reported in a meaningful way.
- One method of reporting the results of a hypothesis test is to report the size, α, of the test used and the decision to reject H_0 or accept H_0.
- Another way of reporting the results of a hypothesis test is to report the value of a certain kind of test statistics, called p-value.

P-value

- A p-value $p(X)$ is a test statistic satisfying $0 \leq p(X) \leq 1$ for every sample point x. Small values of $p(X)$ give evidence that H_1 is true. A p-value is valid if, every $\theta \in \Omega_0$ and every $0 \leq \alpha \leq 1$,

$$P_\theta(p(X) \leq \alpha) \leq \alpha.$$

- If $p(X)$ is a valid p-value, we can easily construct a level α test based on $p(X)$. The test that rejects H_0 if and only if $p(X) \leq \alpha$.
- The p-value is the smallest α at which we would reject H_0. In other words, we reject at all $\alpha \geq p$.
- So, if the p-value is 0.02, then we would reject at $\alpha = 0.05$ but not $\alpha = 0.01$.

Valid P-value

- Theorem 8.3.27: Let $W(X)$ be a test statistic such that large values of W give evidence that H_1 is true. We define

$$p(x) = sup_{\theta \in \Omega_0} P_\theta(W(X) \geq W(x))$$

  Then, $p(X)$ is a valid p-value.
- Proof: Fix $\theta \in \Omega_0$. Let $F_\theta(w)$ be the cdf of $-W(X)$. Define $p_\theta(x) = P_\theta(W(X) \geq W(x)) = P_\theta(-W(X) \leq -W(x)) = F_\theta(-W(x))$.
- Probability integral transformation: X has continuous cdf $F_X(x)$. Let $Y = F_X(X)$. Then, Y is uniformly distributed on (0,1).
- Because $p(x) = sup_{\theta' \in \Omega_0} p_{\theta'}(x) \geq p_\theta(x)$, by the Probability Integral Transformation, we obtain

$$P_\theta(p(X) \leq \alpha) \leq P_\theta(p_\theta(X) \leq \alpha) = \alpha.$$

Calculation of P-value

- Suppose we have a test of the form: reject when $T(X_1, \ldots, X_n) > c$. Then, the p-value is

$$p(x) = sup_{\theta \in \Theta_0} P_\theta(T_n(X_1, \ldots, X_n) \geq T_n(x_1, \ldots, x_n)),$$

  where $x_1, \ldots, x_n$ are the observed data and $X_1, \ldots, X_n \sim P_\theta$.
- Theorem: Under H_0, $p(X) \sim Unif(0, 1)$.
- Note that $p(X)$ is not equal to $P(H_0 \mid X_1, \ldots, X_n)$.

Example 8.3.28 (two-sided normal p-value)

- $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$.
- Consider $H_0 : \mu = \mu_0$ versus $H_1 : \mu \neq \mu_0$
- The LTR rejects H_0 for large values of $W(X) = \mid \bar{X} - \mu_0 \mid /(S/\sqrt{n})$.
- Under H_0, $W(X)$ has a Student's t distribution with $n-1$ degrees of freedom.
- . Thus, the p-value:

$$p(x) = 2P(T_{n-1} \geq \mid \bar{x} - \mu_0 \mid /(s/\sqrt{n})),$$

where T_{n-1} has a Student's distribution with $n-1$ degrees of freedom.

Another Calculation of P-value

- Suppose we have a sufficient statistics $S(X)$ for the model $\{f(x. \mid \theta) : \theta \in \Omega_0\}$, where $H_0 : \theta \in \Theta_0$.
- Let $W(X)$ be a test statistic for which large values give evidence that H_1 is true.
- For each sample point x, define

$$(8.3.10)\ p(x) = P(W(X) > W(x) \mid S = S(x))$$

- As argued in the proof of Theorem 8.3.27, as applied to the conditional distribution of X given $S = s$, we can see that for any $0 \leq \alpha \leq 1$,

$$P(p(X) \leq \alpha \mid S = s) \leq \alpha.$$

Another Calculation of P-value

- Then, for any $\theta \in \Theta_0$, we have

$$P_\theta(p(X) \leq \alpha) = \sum_s P(p(X) \leq \alpha \mid S = s) P_\theta(S = s) \leq$$

$$\sum_s \alpha P_\theta(S = s) \leq \alpha$$

The Permutation Test (Wasserman Chapter 11.5)

- The permutation test is a nonparametric method for testing whether two distribution are the same.
- This test does not involve any asymptotic approximations.
- Suppose we have data $X_1, \ldots, X_n \sim F$ and $Y_1, \ldots, Y_m \sim G$.
- Our null and alternative hypotheses:

$$H_0 : F = G \text{ versus } H_1 : F \neq G.$$

- Let $Z = (X_1, \ldots, X_n, Y_1, \ldots, Y_m)$.
- Creat labels $L = (1, \ldots, 1, 2 \ldots, 2)$ with the number of 1 being n and the number of 2 being m.
- A test statistic can be written as a function of Z and L.
- $T = g(L, Z)$ be some test statistics.

The Permutation Test (Wasserman Chapter 11.5

- For example,

$$T = \left| \frac{\sum_{i=1}^{N} Z_i I(L_i = 1)}{\sum_{i=1}^{N} I(L_i = 1)} - \frac{\sum_{i=1}^{N} Z_i I(L_i = 2)}{\sum_{i=1}^{N} I(L_i = 2)} \right|,$$

  where $N = m + n$. Then,

$$T = \left| \bar{X}_n - \bar{Y}_m \right| .$$

- For each permutation, compute the test statistic T . Denote these values by $T_1, \ldots, T_{N!}$. If the null hypothesis is true, each of these values is equally likely.
- Let t_{obs} be the observed value of T. Assuming rejecting H_0 if T is large, the p-value is given by $p - value = P_0(T > t_{obs}) = \frac{1}{N!} \sum_{j=1}^{N!} I_{(T_j > t_{obs})}$.

The Permutation Test

- Or, we can calculate the p-value as follows: $p = \frac{1}{N!} \sum_{\pi} I(g(L_\pi) > g(L, Z))$, where L_π is a permutation of the labels and the sum is ver all permutations.
- Toy Example: observed data $(X_1, X_2, Y_1) = (1, 9, 3)$. Let $T = \mid \bar{X} - \bar{Y} \mid$.

permuation	value of T	probability
(1,9,3)	2	1/6
(9,1,3)	2	1/6
(1,3,9)	7	1/6
(3,1,9)	7	1/6
(3,9,1)	5	1/6
(9,3,1)	5	1/6

P-value$= P(T > 2) = 4/6$.

The Permutation Test

Summing over all permutations is infeasible. But it is sufficient to use a random sample of permutations. So we do this:

- Compute a random permutation of the labels and compute W .
- Do this K times giving values $T^{(1)}, \ldots, T^{(K)}$.
- Compute the p-value

$$\frac{1}{K}\sum_{j=1}^{K} I(T^{(j)} > T).$$

First Permutation Test

- Fisher's famous Lady Tasting Tea experiment (Salsburg, 2001)
- In one summer afternoon in Cambridge in the 1920's, a group of people, including Ronald Aylmer Fisher, sit outside a table for afternoon tea, and one of the women was insisting that tea tasted different depending upon whether the tea was poured into the milk or whether the milk was poured into the tea.
- Fisher designed and conducted a randomized experiment to test the woman's claim, along with a detailed description of how to conduct a simple experiment to test the claimed ability, and how to obtain an exact solution. This detail explanation was given in Chapter 2 of Fisher's book, entitled "Statistical Methods for Research Workers".

Lady Tasting Tea



Fisher's Experiment:

- Mixing 8 cups of tea, with four milk first and other four tea first.
- Giving each cup to the lady in random order for judgement.
- The lady knew that there were 4 cups of each mixture order.

Lady Tasting Tea



		True Order		Total
		Tea First.	Milk First	
Lady's guesses	Tea First	3	1	4
	Milk First	1	3	4
Total		4	4	8

Permutation Test

- Under the null hypothesis that the Lady does not have the ability of telling whether the milk or the tea infusion was first added the cup, we have $\binom{8}{4} = \frac{4!4!}{8!} = 70$ possible orderings of these 8 cups, and the lady would equally likely chose each of the total of 70 outcomes.
- The lady gave 6 correct guesses, and among 70 possibles guesses, there are 17 outcomes with 6 or more (out of 8) correct guesses, yielding a p-value of $17/70 = 0.24286$.
- Recall the definition of p-value: what is the likely chance for the lady to have her result at least as extreme as the one observed.